

JAYANTHAN SENTHILKUMAR

Software Engineer | Backend Developer | AI & Machine Learning Engineer

+91 8825756388 | jayanthansenthilkumar18@gmail.com | jayanthan.fyi

linkedin.com/in/jayanthan18 | github.com/jayanthansenthilkumar | Karur, Tamil Nadu, India

PROFESSIONAL SUMMARY

Software Engineer experienced in building distributed backend systems, RESTful APIs, and full-stack web apps using Python, PHP, Laravel, and MySQL cutting backend latency by 30% and boosting API reliability by 40% as an industrial trainee. Built end-to-end projects in AI-assisted trip optimization, IoT fleet management, and enterprise ERP. Bachelor of Technology in AI & Machine Learning, CGPA 8.08.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, PHP, Java, HTML, CSS

Frameworks: Laravel, Flask

Databases: MySQL (query optimization, indexing, schema design), SQLite

Enterprise Technologies: SAP ABAP Cloud, SAP BTP, ABAP RAP, SAP HANA Cloud

AI & Machine Learning: Predictive Modelling, Applied ML-Assisted Feature Design, Data-Driven Decision Logic

Developer Tools: Git, GitHub, GitHub Actions (CI/CD), IoT Integration (Arduino)

PROFESSIONAL EXPERIENCE

Technology Innovation Hub, MKCE — Software Developer Trainee

Apr 2024 – Jul 2025

- Engineered a distributed backend system with PHP and SQL, reducing service latency by 30% and boosting throughput for a growing user base.
- Designed RESTful APIs standardizing inter-service communication, improving system reliability by 40% and cutting integration errors.
- Optimized MySQL queries and indexing strategy, cutting data processing time by 35% and improving responsiveness under load.

PROJECTS

Ishaa - Python-Based AI Framework | Python, ASGI, SQLite, WebSockets, GraphQL, OpenAPI, JWT, REST APIs, CLI, GitHub Actions

- Architected a zero-dependency ASGI-native Python framework with custom ORM, JWT auth, WebSockets, and GraphQL, cutting web app development effort by 60%.
- Engineered proprietary routing, execution, and quality-pipeline modules for intelligent routing and automated CI/CD evolution, improving performance and deployment reliability by 45%.

KR Connect - Smart College ERP | PHP, MySQL, Ajax, jQuery, HTML, CSS, JavaScript

- Developed Discipline and Grievance Management modules using PHP, Ajax, and MySQL, streamlining issue tracking and record updates by 50% for college administration.
- Built real-time administrative dashboards using HTML, CSS, and JavaScript, improving page load speed and operational efficiency by 45%.

PLANGO - AI-Assisted Trip Planner | Flask, Python, SQLite, JavaScript

- Built a trip-optimization web application ingesting real-time traffic data to generate route recommendations, improving route accuracy by 40% and reducing computed route latency by 60%.
- Designed predictive logic for traffic and travel-time estimation, contributing to a 35% increase in measured user satisfaction during testing.

EDUCATION

Bachelor of Technology, Artificial Intelligence and Machine Learning

2022 – 2026

M. Kumarasamy College of Engineering, Karur, India | CGPA: 8.08/10

CERTIFICATIONS

- GitHub Foundations - GitHub | SAP – ABAP Backend Development - SAP
- Google Play & Unity - Associate Game Development - Unity Technologies

LEADERSHIP & ACHIEVEMENTS

- Finalist, PECHACKS 2.0 Hackathon
- Guest Lecturer, "Git vs GitHub" Workshop - Panimalar Engineering College, Chennai
- Recipient, Student of the Month - Technical Leadership & Performance